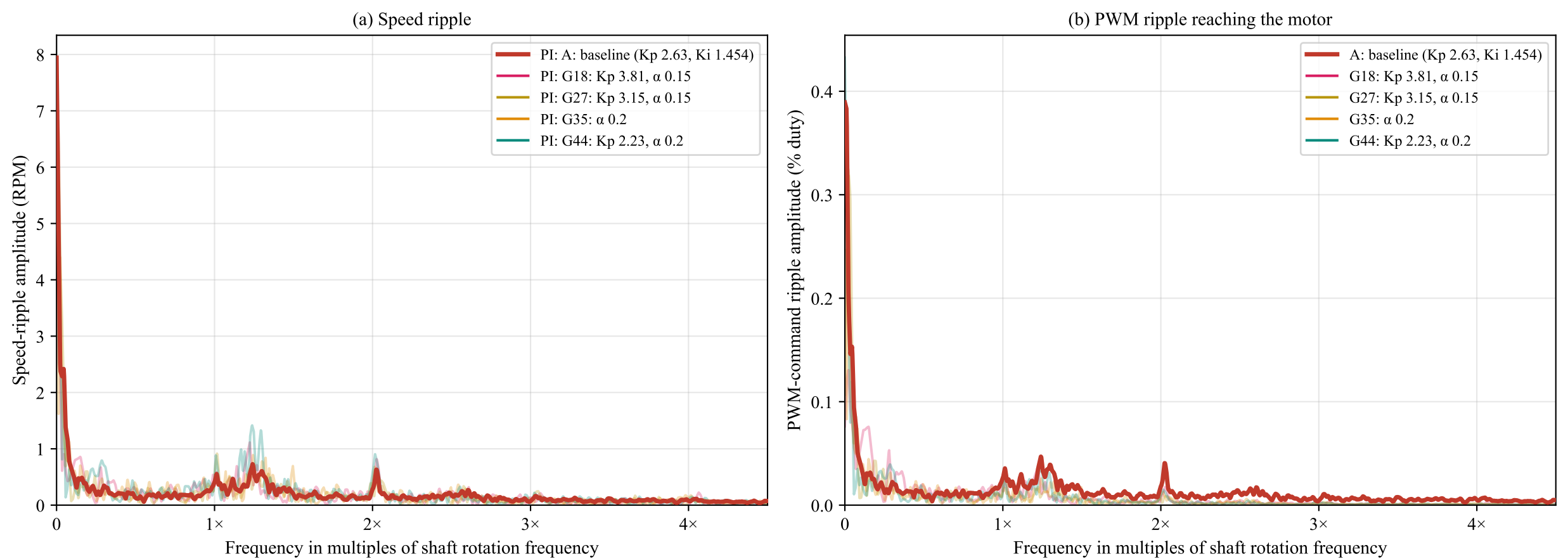

(a) Speed ripple
Speed-ripple amplitude (RPM)
Frequency in multiples of shaft rotation frequency
PI: A: baseline (Kp 2.63, Ki 1.454)
PI: G18: Kp 3.81, α 0.15
PI: G27: Kp 3.15, α 0.15
PI: G35: α 0.2
PI: G44: Kp 2.23, α 0.2
0
1
2
3
4
5
6
7
8
0
1×
2×
3×
4×
(b) PWM ripple reaching the motor
PWM-command ripple amplitude (% duty)
Frequency in multiples of shaft rotation frequency
A: baseline (Kp 2.63, Ki 1.454)
G18: Kp 3.81, α 0.15
G27: Kp 3.15, α 0.15
G35: α 0.2
G44: Kp 2.23, α 0.2
0.0
0.1
0.2
0.3
0.4
0
1×
2×
3×
4×